



Research article

Artificial intelligence-based power market price prediction in smart renewable energy systems: Combining prophet and transformer models

Cheng Huang^a, Tianhui Zhao^a, Di Huang^b, Bingcheng Cen^a, Qian Zhou^a, Wenjia Chen^a

^a State Grid Jiangsu Electric Power Co., LTD, Electric Power Research Institute, 211103, Nanjing, Jiangsu, China

^b State Grid Jiangsu Electric Power Co., Ltd, Double Innovation Center, 210024, Nanjing, Jiangsu, China



ARTICLE INFO

Keywords:

Renewable energy systems
Deep learning
Electricity market
Price forecasting,
Multi-source data
Data analysis
Artificial intelligence

ABSTRACT

With the increasing integration of smart renewable energy systems and power electronic converters, electricity market price prediction is particularly important. It is not only crucial for the interests of power suppliers and market regulators but also plays a key role in ensuring the reliable and flexible operation of the power system, particularly during extreme weather events or abnormal conditions. This study develops a hybrid time series forecasting model that combines Prophet and Transformer, which takes advantage of deep learning to provide a new solution for electricity market price forecasting. By introducing the Stacking optimization strategy, this study improves the accuracy and stability of electricity market price sequence prediction. In addition, the study tries to integrate traditional time series forecasting methods (such as the Prophet model) with deep learning models (such as the Transformer model), aiming to make full use of their respective advantages to achieve more accurate and stable predictions. Through experimental evaluation on four electricity market data sets, this study finds that the hybrid forecast model exhibits significant performance improvements in enhancing the accuracy and stability of electricity market price predictions. This method not only provides a more accurate tool for power market price prediction, but also provides solid technical support for the efficient operation and sustainable development of smart renewable energy systems. Experimental results also show that the combination of deep learning models with traditional time series methods and the introduction of Stacking strategies is crucial to improving the performance of power market price prediction, and also helps us better understand and design smart renewable energy systems, price and energy management strategies, thereby providing an effective method for achieving efficient and reliable power and energy transmission.

1. Introduction

With the transformation of the global energy structure, smart renewable energy systems have gradually become a core component of the power system. In this context, accurate electricity market price forecasting is of great significance for maintaining the balance

E-mail addresses: HCheng921@163.com (C. Huang), eethzhao@163.com (T. Zhao), andy_r1@msn.com (D. Huang), cbc_whu@126.com (B. Cen), xjtu@163.com (Q. Zhou), 921131@163.com (W. Chen).

<https://doi.org/10.1016/j.heliyon.2024.e38227>

Received 26 January 2024; Received in revised form 18 September 2024; Accepted 20 September 2024

Available online 18 October 2024

2405-8440/© 2024 Published by Elsevier Ltd.

(<http://creativecommons.org/licenses/by-nc-nd/4.0/>).

This is an open access article under the CC BY-NC-ND license

of energy supply and demand, optimizing grid operation, and promoting the effective use of renewable energy [1]. However, due to the volatility of renewable energy such as wind and solar energy, electricity market prices often present a high degree of complexity and uncertainty, and traditional forecasting methods face many challenges in dealing with these dynamic changes [2,3]. Therefore, how to build a more accurate and stable forecasting model has become a hot issue in current research [4].

The electricity market price forecasting not only needs to consider the historical data of the market, but also must integrate multi-source heterogeneous data such as weather and power generation [5]. Although traditional time series models can process one-dimensional time data, they often fail to achieve ideal results when faced with complex multi-dimensional features and long sequence dependencies [6]. In order to solve these problems, the application of artificial intelligence technology, especially deep learning models, has become an important breakthrough [7].

This paper proposes a model framework combining Prophet and Transformer for the forecasting and trend analysis of electricity market prices. The Prophet model is good at capturing long-term trends and seasonal fluctuations in time series, especially when dealing with exogenous events such as holidays [8,9]. On the other hand, the Transformer model has significant advantages in modeling complex time series dependencies and can effectively capture the nonlinear dynamic characteristics of price data [10]. However, relying on a single model may have certain limitations, so this article further introduces the Stacking model to optimize model performance and improve the final prediction accuracy by integrating the output of the Prophet and Transformer models.

The key contributions of this research can be summarized as follows:

- An innovative hybrid time series forecasting model is proposed, which combines the Prophet and Transformer models, and introduces the Stacking optimization strategy to solve the power market price forecasting problem.
- Wider scope of application. This study focuses on the field of smart grid, and the proposed model can not only be used for electricity market price prediction, but also suitable for other smart grid tasks, such as power load type prediction.
- Data efficiency and interpretability are strong. Compared with some complex deep learning models, the model in this study has higher data efficiency while maintaining a certain degree of interpretability.

2. Related work

2.1. Deep learning in smart grid monitoring

As an important part of modern power system, the research focus of smart grid is on how to improve the stability and flexibility of power grid through efficient data processing and intelligent decision-making [11,12]. In smart grid, CNN and GCN are gradually applied to the state monitoring and prediction of power system. CNN performs well in processing images and time series data in power system, especially in identifying complex spatial features. For example, CNN is very successful in the application of power equipment fault detection, which can quickly identify abnormal signals from monitoring images [13]. However, CNN is subject to certain limitations when processing non-image structured data, and it is difficult to capture the topological relationship in the power network [14,15].

In contrast, GCN can effectively model complex power transmission networks in smart grids by utilizing the network structure of the power system. For example, in power load forecasting and power grid state estimation, GCN is widely used to capture complex dependencies between power nodes [16]. The advantage of this method is that it can make full use of the graph structure characteristics of the power grid and improve the prediction accuracy of the model. However, GCN performs worse than time series models when dealing with long time series dependencies [17]. Therefore, it is often necessary to combine other time series models, such as LSTM and Seq2Seq, in the dynamic prediction of smart grids to enhance its time series modeling capabilities.

2.2. Electricity price forecasting with deep learning

Electricity market price prediction is an important topic in smart grid research. In recent years, time series models based on deep learning have made significant progress in this field [18]. LSTM and Seq2Seq are commonly used models in price prediction. LSTM is good at processing time series data with long-term dependencies, and can effectively capture long-term and short-term trends in historical prices through memory and forgetting mechanisms. In practical applications, LSTM is widely used to predict electricity market prices and can better cope with complex fluctuations in the electricity market. However, LSTM has limited flexibility when faced with multi-dimensional and multi-variable data [19,20].

The Seq2Seq model is particularly suitable for multi-step prediction tasks. The application of Seq2Seq in electricity market price prediction improves the modeling ability of long-term series dependencies by introducing an attention mechanism [21]. For example, some studies use Seq2Seq combined with the Transformer model to show higher accuracy and stability in multi-step prediction tasks of electricity market prices [22]. However, the Seq2Seq model relies heavily on data during the training process, takes a long time to train, and has high model complexity, which may pose challenges to scenarios with limited computing resources.

In order to further improve the accuracy of electricity price prediction, Bidirectional Gated Recurrent Units (BiGRU) have also been introduced into price prediction tasks in recent years. BiGRU can handle both forward and reverse time dependencies, and has certain advantages in capturing complex timing relationships [23,24]. However, BiGRU still needs to be improved when dealing with multi-source data fusion and handling long-term time dependencies, so it is often used in combination with other models to improve prediction effects.

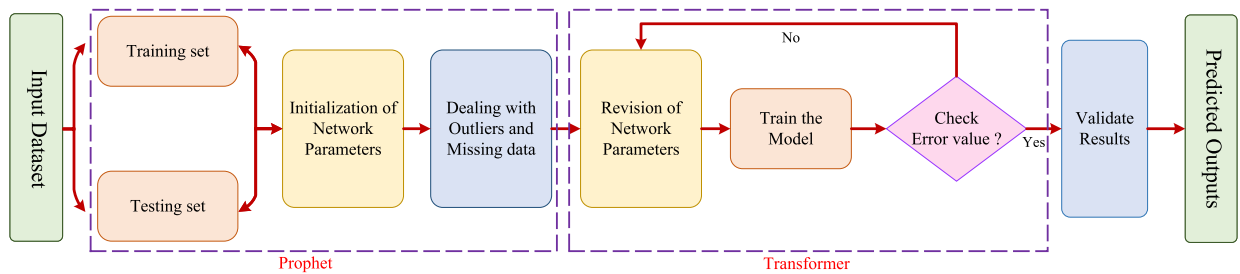


Fig. 1. Overall flow chart of the model. The Prophet module is used to initialize network parameters and process abnormal data, while the Transformer module further performs time series modeling and optimizes parameter settings, and finally outputs the forecast results through error checking and result verification.

2.3. Deep reinforcement learning in power systems

Deep reinforcement learning (DRL), as a technology that has emerged in recent years, has been increasingly used in smart grids and power markets [25,26]. Typical DRL applications include optimizing power dispatch through DQNs to improve the utilization efficiency of renewable energy. For example, the application of DRL in power load management and real-time pricing can dynamically adjust power prices according to market supply and demand conditions to reduce fluctuations and improve market efficiency [27,28].

DRL performs poorly when processing high-dimensional continuous action spaces and is prone to falling into local optimal solutions [29]. In order to cope with these problems, researchers have proposed hybrid models combined with deep learning, such as combining DRL with Transformer for complex power market pricing optimization, thereby improving prediction accuracy and accelerating model convergence [30]. However, the generalization ability and stability of DRL are still the focus of current research. How to build a more robust DRL model in a changing power market environment is one of the current important challenges.

Various deep learning models such as CNN, GCN, LSTM, Seq2Seq, BiGRU and DRL have shown different advantages and disadvantages in related applications. However, when faced with a complex power market environment, these models still have problems such as insufficient fusion of multi-source heterogeneous data, inaccurate modeling of long sequence dependencies, and high model training costs. The Prophet-Transformer model proposed in this paper, by introducing Stacking optimization, can not only combine Prophet to capture basic trends and seasonal characteristics, but also use Transformer and Stacking to improve the modeling capabilities of complex time series dependencies, thereby providing a more accurate and stable solution for power market price forecasting.

3. Methodology

3.1. Overview of the network

This paper proposes a power market price forecasting model that combines Prophet and Transformer. The model is processed in two stages: first, the Prophet model is used to capture the long-term trends and seasonal fluctuations in the power market, and then the Transformer model is used to model complex time series dependencies, thereby improving the accuracy and stability of the forecast. In order to further improve the performance of the model, this paper also introduces the Stacking model to integrate the outputs of Prophet and Transformer to optimize the final forecasting effect.

As shown in Fig. 1, the overall process of the model starts with the input data set, including the division of training and test sets. First, the Prophet model is used to initialize network parameters and process outliers and missing data in the data. The Prophet model is good at capturing long-term trends and seasonal characteristics in the data, ensuring that a relatively stable basic feature is provided for the Transformer in the initial stage. Based on the data processed by the Prophet model, the Transformer model further corrects the network parameters and performs complex time series modeling. The Transformer model can capture short-term fluctuations and long-term series dependencies in the power market through its powerful self-attention mechanism. During the model training process, the network parameters are dynamically adjusted until the error value of the model meets the requirements. Finally, the output of the model will be verified and the forecast results of the power market price will be generated. The two are effectively combined through the Stacking optimization module to form a multi-model integrated forecasting system, which improves the accuracy and robustness of the forecast.

The detailed steps for running the Prophet-Transformer model are described in Algorithm 1.

3.2. Prophet module

The Prophet model is a powerful tool for time series data prediction proposed by Facebook. It is especially good at processing time series data with significant trends, periodicity and abnormal events [31]. Prophet models provide accurate forecasts for future points in time by modeling long-term trends, seasonal changes, and holiday effects in time series. In electricity market price forecasting, market prices often have significant cyclical characteristics, such as day and night price fluctuations and seasonal demand changes. Therefore, this paper selects the Prophet model as the preliminary time series processing module responsible for capturing these long-term and cyclical trends.

Algorithm 1 Prophet-Transformer Training.**Input:**

- Training dataset: D with datasets - ["ENTSO-E Database", "PJM Electricity Market Database", "CAISO Database", "ERCOT Database"]
- Prophet-Transformer model: $\mathcal{M}$

Output:

- Trained Prophet-Transformer model: $\mathcal{M}$

Initialization: Initialize model parameters and hyperparametersDefine loss function $\mathcal{L}$, optimization algorithm (e.g., Adam)

Set convergence criteria

Initialize evaluation metrics: Recall, Precision

while *Not converged* **do** Sample mini-batch from D Initialize gradients: $\nabla\theta \leftarrow 0$ **for each** sequence X **in the mini-batch** **do** Encode input sequence using $\mathcal{M}$ Predict future values: $\hat{Y} = \mathcal{M}(X)$ Compute loss: $\mathcal{L}_X = \mathcal{L}(\hat{Y}, Y_{\text{true}})$ Compute gradients: $\nabla\theta_X = \nabla\theta_X + \frac{\partial\mathcal{L}_X}{\partial\theta}$ **end** Update model parameters: $\theta \leftarrow \theta - \text{optimizer}(\nabla\theta)$ **if** *Convergence criteria met* **then** **break** **end****end****Evaluation:** **while** *Not converged* **do** Sample validation dataset: $\mathcal{V}$

Initialize evaluation metrics: Recall, Precision

for each sequence X **in** $\mathcal{V}$ **do** Encode input sequence using $\mathcal{M}$ Predict future values: $\hat{Y} = \mathcal{M}(X)$ Compute evaluation metrics based on $\hat{Y}$ and Y_{true} **end**

Calculate average evaluation metrics

if *Average evaluation metrics meet stopping criteria* **then**

Model converged

break **end****end****Return:** Trained Prophet-Transformer model $\mathcal{M}$

The main advantages of the Prophet model are its robustness to data outliers and flexible interpretability. Electricity market data are often affected by abnormal events (such as equipment failures, extreme weather), and the Prophet model can well identify these anomalies and correct them by adjusting model parameters [32]. In addition, the Prophet model has strong interpretability, and the model can clearly give the specific impact of trends, seasonality and holiday effects, which is important for analyzing the causes of price fluctuations in the power market.

In the overall framework of this article, the main role of the Prophet module is to provide processed preliminary prediction results for the Transformer module. Through the Prophet model, the system can effectively model long-term trends and seasonal fluctuations in market prices, and the Transformer module further captures complex time series dependencies in price data. The two work together, with the Prophet model responsible for providing stable trend predictions at a global level, while the Transformer model models short-term fluctuations in detail.

As can be seen from Fig. 2, the processing flow of the Prophet model starts from the SG data source. After the data is obtained by the smart meter, it is filtered and preprocessed, which mainly includes processing missing data and abnormal values. The Prophet model performs model fitting on the cleaned data. This process mainly decomposes and reconstructs the data through the trend and seasonal components of the Prophet model. The fitted model can generate time series prediction results, which will be used as preliminary predictions in subsequent model algorithms.

The Prophet module effectively decomposes time series data through the following formula, captures the impact of its trends, seasonality, and outliers, and provides robust initial forecast results for subsequent Transformer models.

$$g(t) = k + mt \quad (1)$$

where k represents the intercept, m represents the trend slope, and t is the time variable. This function is used to capture long-term linear trends in time series.

$$s(t) = \sum_{i=1}^N \left(a_i \cos\left(\frac{2\pi it}{T}\right) + b_i \sin\left(\frac{2\pi it}{T}\right) \right) \quad (2)$$

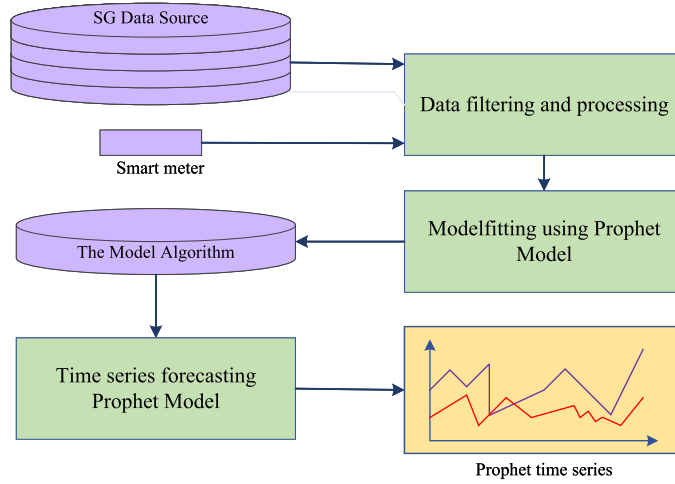


Fig. 2. Structural flow chart of the Prophet module.

where a_i and b_i are model parameters respectively, T is the period, and N is the number of seasonal components. This function is used to capture cyclical fluctuations in time series.

$$h(t) = \sum_{k=1}^K \left(\beta_k 1_{\{t \in H_k\}} \right) \quad (3)$$

where H_k represents the k th holiday, and β_k is the intensity of the holiday effect. This function is used to handle the impact of holidays or special events on time series.

3.3. Transformer model

Unlike traditional RNN or LSTM models, Transformer does not rely on sequence order, but uses a multi-head attention mechanism to capture global dependencies in the data [33]. This enables it to have stronger parallel computing capabilities and faster training speed when processing long-term series dependencies. This paper introduces the Transformer module for power market price forecasting, aiming to further capture the complex short-term fluctuations and time series dependencies in the data [34].

The Transformer module plays a key role in the overall model of this paper. Transformer uses its global attention mechanism to capture the complex dependencies between various time points in the power market, especially when dealing with long-term dependencies and nonlinear fluctuations. In addition, Transformer's multi-head attention mechanism and position encoding enable it to efficiently model multi-dimensional feature data and provide more accurate price predictions.

As illustrated in Fig. 3, the Transformer module receives time series data with positional encoding, which preserves the temporal information of the data. The core architecture of the Transformer is composed of an encoder and decoder, where each component integrates a multi-head attention mechanism and a feed forward network. To maintain model stability and efficiency, the "Add & Norm" operation is applied throughout the process.

Within the encoder, the input data initially flows through the multi-head attention mechanism. This mechanism processes the data in parallel using several attention heads, allowing it to identify relationships in the time series from various perspectives.

$$\text{Attention}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (4)$$

Where Q is the query matrix, K is the key matrix, V is the value matrix, and d_k is the scaling factor. This formula is used to calculate the correlation between different time steps in the input sequence, and the output is the weighted features of the importance of each time step.

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h) W^O \quad (5)$$

Where, $\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$, W^O is the projection matrix. The multi-head attention mechanism works in parallel with multiple attention heads and can learn multi-dimensional correlation features from different subspaces.

After the encoder is processed, the data enters the decoder. The decoder works similarly to the encoder, but introduces the masked multi-head attention mechanism to ensure that only the current and previous time step information can be used in sequence prediction to avoid information leakage. This process is crucial for time series prediction because future data should not be used in advance. The final output of the decoder passes through the fully connected layer and the Softmax layer to generate the probability distribution of price prediction.

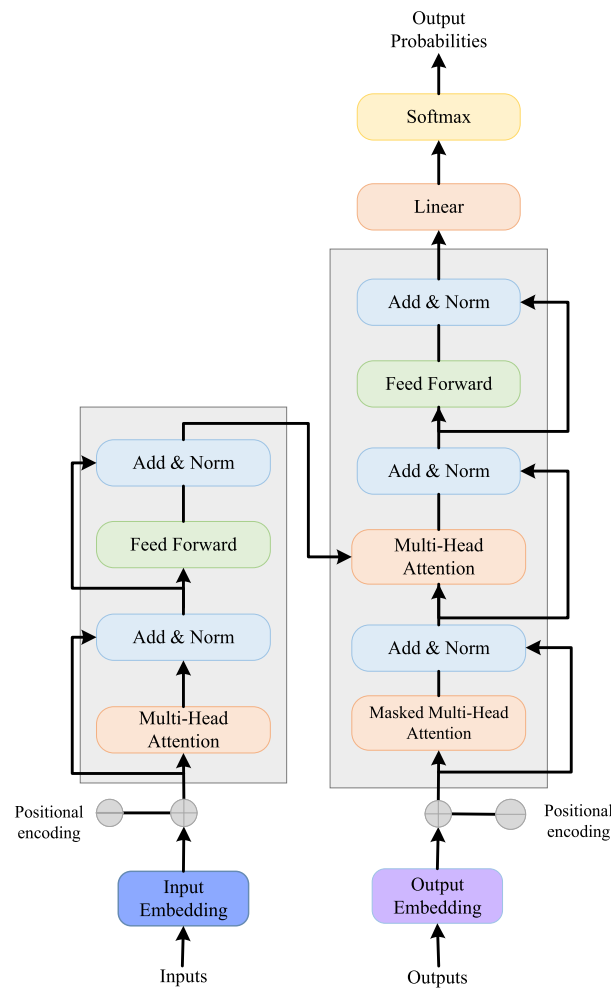


Fig. 3. Structural flow chart of the Transformer model.

This module works with the Prophet module. Prophet is responsible for modeling long-term trends and seasonal changes, while Transformer focuses on short-term fluctuations and global dependencies. The combination of the two effectively improves the accuracy and robustness of the prediction. Especially when dealing with multi-source heterogeneous data, the introduction of the Transformer module significantly improves the model's time series modeling capabilities. The Transformer module plays an important role in the electricity market price forecasting task in this paper through its powerful global modeling capabilities and flexible attention mechanism, especially providing significant performance improvements in capturing complex short-term fluctuations and time series dependencies.

3.4. Stacking optimization model

The Stacking optimization model is an integral part of the Prophet-Transformer combination model. It is mainly responsible for improving the accuracy and robustness of electricity market price prediction [35]. The core of this process is to utilize various data characteristics and patterns captured by different models and effectively fuse this information to form the final prediction. Within the model, Stacking works by integrating the respective outputs of the Prophet and Transformer models [36].

The Prophet model focuses on capturing trends and cyclical changes in price data, while the Transformer model is good at capturing complex serial dependencies, such as short-term fluctuations and non-linear characteristics. The Stacking model treats these outputs as different feature combinations and gives them different importance through the weight parameters obtained through training, so that the final prediction result can not only reflect long-term trends, but also respond quickly to instant changes in the market.

The core advantage of the Stacking model lies in its synthesis and optimization of predictions from different sources. By assigning appropriate weights to each model, Stacking can balance between various models, reduce the bias that a single model may introduce, and at the same time improve the generalization ability of the overall prediction system. This approach enables the system to maintain robust performance and optimize prediction results when faced with different data sets and market conditions. In order to ensure the efficiency and adaptability of the Stacking process, this article carefully designs the interaction of each component. According

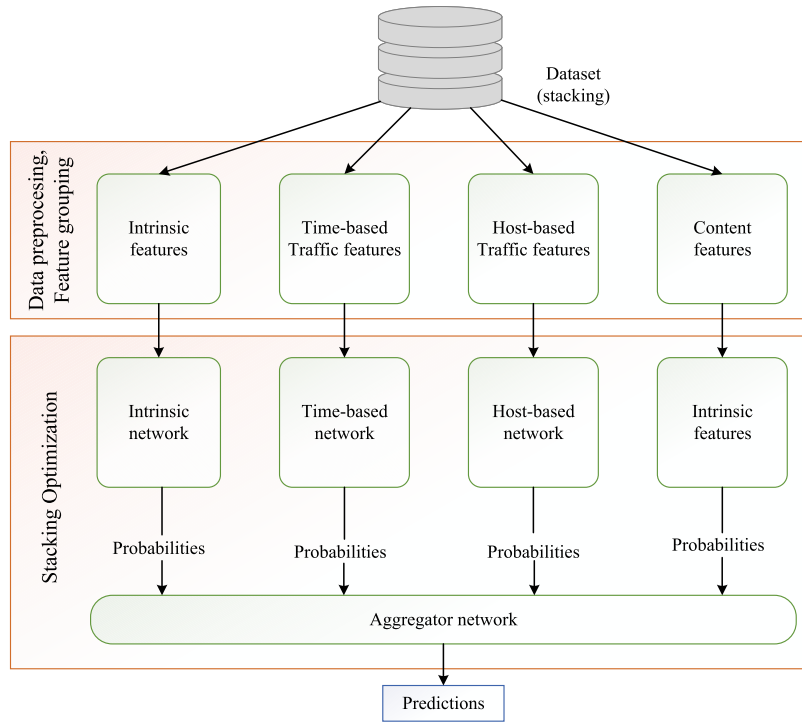


Fig. 4. Structural flow chart of the Stacking model.

to the performance of different models on specific data sets, the weight parameters are dynamically adjusted through the adaptive mechanism, which improves the adaptability of the model when facing unknown data and also enhances the prediction stability in various market fluctuations.

The operation process of Stacking Optimization Model is shown in Fig. 4.

The main formula of Stacking Optimization Model is as follows:

$$\hat{y}_{\text{stacking}} = \alpha_1 \hat{y}_{\text{Prophet}} + \alpha_2 \hat{y}_{\text{Transformer}} \quad (6)$$

among them, $\hat{y}_{\text{stacking}}$ represents the final prediction output of the Stacking model, which is the price prediction result integrating the output of the Prophet and Transformer models. $\hat{y}_{\text{Prophet}}$ represents the price prediction output of the Prophet model. $\hat{y}_{\text{Transformer}}$ represents the price prediction output of the Transformer model. α_1 and α_2 represent weight parameters, which are used to control the relative importance of each model output. These parameters can be determined through training and optimization to achieve the best integration effect.

4. Experiment

4.1. Datasets

This paper mainly uses the following four data sets to study the price prediction and trend analysis problems of smart grid power market.

The ENTSO-E database is a data resource maintained by the European Association of Electricity Transmission System Operators (ENTSO-E) to support power system monitoring, operations and research. The ENTSO-E database includes millions of electricity market price time series data points. Data includes timestamps, market prices, load data, generation information and related market operation data. The data sources of this database mainly include power system operators in European countries, as well as relevant market and regulatory agencies. The ENTSO-E database provides a wide range of power system data, including power load, power generation, transmission line status, market prices, and market operation data. Data sets span multiple years and often include historical and real-time data. Historical data dating back decades is critical for long-term analysis and planning of power systems. Real-time data is updated in real-time and used for actual monitoring and operation of the power system [37]. This broad time span makes the ENTSO-E database a valuable resource for studying electricity market price forecasting, load forecasting, and power system behavior. The ENTSO-E database covers multiple countries and regions within the European continent. This includes members of the European Joint Electricity Market, such as Germany, France, Spain, etc., as well as other countries interconnected with the European electricity system.

The PJM Electricity Market Database is a data resource maintained by PJM Interconnected Power Company in the United States and is used to support electricity market monitoring, operations, and research. The PJM Electricity Market Database contains millions of time series data points. Data includes market prices, electricity loads, transmission line status and market operations data. The PJM electricity market covers the Midwestern region of the United States, including 13 states and DC. The database's data sources mainly include PJM Interconnected Power Companies, power system operators, and relevant market and regulatory agencies [38]. The time span of the PJM electricity market database covers multiple years and typically includes historical and real-time data. Historical data, stretching back many years, is critical for long-term analysis and planning of electricity markets. Real-time data is updated at intervals of minutes or less and is used for real-time monitoring and operation of the power system. Its geographical scope covers the Midwestern region of the United States, including Pennsylvania, New Jersey, Illinois, Ohio and other places. The electricity markets in these regions are interconnected, forming a complex electricity system. The PJM power market database is commonly used to support power market price forecasting, power load forecasting, market operation analysis, and power system planning research.

The CAISO database is a data resource maintained by the California Independent System Operator (CAISO) to support power system monitoring, operations, and research. The CAISO database includes millions of time series data points. Includes market prices, electricity loads, transmission line status and market operations data. CAISO manages California's electric power system, including the electricity market and transmission network. The data sources of this database mainly include CAISO itself, power system operators, and relevant market and regulatory agencies. The time span of the CAISO database is very wide, including historical data and real-time data [39]. Historical data covers many years and is critical for long-term planning and analysis of power systems. Real-time data is updated at intervals of minutes or less and is used for real-time monitoring and operation of power systems. This time span makes the CAISO database an important source of data on electricity market prices, electricity load and generation. The CAISO database geographically covers all of California. In research, the CAISO database is usually used to support power market price forecasting, power load forecasting, market operation analysis, and power system planning research.

The ERCOT database is a data resource maintained by the Electric Reliability Council of Texas (ERCOT) to support power system monitoring, operations, and research. ERCOT manages Texas' electricity system, including the electricity market and transmission network. The ERCOT database includes millions of time series data points. Data includes market prices, electricity loads, transmission line status and market operations data. The database's data sources mainly include ERCOT itself, power system operators, and relevant market and regulatory agencies. The time span of the ERCOT database is also very wide, including historical data and real-time data. Historical data goes back many years, and real-time data is updated at intervals of minutes or less for real-time monitoring and operation of power systems. This time span makes the ERCOT database an important source of data on electricity market prices, electricity load and generation [40]. In terms of geographic scope, the ERCOT database covers all of Texas. Texas is one of the largest states in the United States with a large electric power system and market. The ERCOT database is usually used to support power market price forecasting, power load forecasting, market operation analysis, and power system planning research.

The above electricity market data are particularly suitable for prediction by the Prophet-Transformer model, mainly because they have the following key characteristics: First, time dependence, the electricity market price data has obvious time series characteristics, such as cyclical changes (Within days, weeks, seasonal changes) and trend changes, these are the advantages of the Prophet model. Then there is high volatility. Electricity prices are affected by many factors, such as weather conditions, policy changes, supply and demand, etc., resulting in high price volatility. The self-attention mechanism of the Transformer model can effectively capture these short-term nonlinear changes. Finally, there is multi-source data integration. Electricity market data usually comes from multiple sources, such as real-time consumption data, weather forecasts, historical prices, etc., which requires the model to be able to process and integrate multiple types of input data. Transformer models have natural advantages in processing multi-source inputs.

4.2. Experimental setup

This study combines the Prophet and Transformer models and introduces the Stacking optimization strategy to study the price prediction and trend analysis problems of the smart grid power market. To ensure accuracy and reproducibility, experimental details need to be carefully designed. The experimental setup and details are as follows:

Step 1: Dataset preparation

- Data source: Research group obtained electricity market price time series data, covering the time span from 2010 to 2023, and from four databases, ENTSO-E, PJM, CAISO and ERCOT, covering market data in different geographical ranges.
- Data preprocessing: Research group cleaned the original data and dealt with missing values, outliers and outliers. At the same time, we normalized the data to ensure comparability between different data sets. For outlier handling, we used the Z-Score method to detect and remove outliers. For missing data, we employed various imputation methods. For small continuous missing data points, we used linear interpolation, which fills in missing values based on the values of adjacent data points before and after the gaps. For larger ranges of missing data, we used time series pattern-based imputation methods, such as filling in missing values with data from similar time points in historical cycles.
- Developed predictor feature description: Weather data includes temperature, humidity, wind speed, etc. Seasonal information includes season, month, and day of the week. Historical price data includes the electricity market price of the previous day or week. Other related market factors include power load, power generation, etc.
- Data set division: Research group divide the data set in chronological order, dividing the data into 70% for training, 15% for validation, and 15% for testing to ensure the generalization performance of the model on future data.

Step 2: Model selection and hyperparameter tuning

- **Model selection:** Research group chose two different types of models, Prophet and Transformer. Prophet is used to capture basic trends and seasonality, while Transformer is used to handle complex sequence relationships.
- **Hyperparameter setting:** For the Prophet and Transformer models, we performed hyperparameter adjustments, including seasonal settings, learning rate, number of layers, number of heads, etc., to optimize model performance. Seasonal components include annual seasonality and weekly seasonality, seasonal parameters include seasonal intensity and seasonal window size, and trend parameters include growth rate. The window size of seasonality is set to 10, the seasonal intensity is set to 0.5, and the growth rate is set to 0.03. The learning rate is set to 0.001, the number of layers is set to 6, the number of heads is set to 8, the maximum sequence length is set to 128, and the embedding dimension is set to 64.

Step 3: Model training process

- **Data preparation:** Research group divide the data into training set, validation set and test set to ensure that the model can generalize. At the same time, we use the sliding window method to obtain sequence data for training of the Transformer model.
- **Prophet model training:** First, Research group prepare training data, including timestamps and target variables (electricity market prices), and then add holiday information. During the training process, hyperparameter adjustments are performed and the validation set is used to evaluate model performance.
- **Transformer model training:** Research group prepare time series data, use the training data to train the Transformer model, and use the validation set to evaluate the performance. Techniques such as batch training and gradient clipping are adopted to ensure the stability and convergence of the model.

Step 4: Loss function and evaluation metrics

- **Loss function:** Considering the continuous nature of electricity market price prediction, this paper uses the MSE as the main loss function. MSE helps directly optimize the squared difference between predicted and actual values, which is especially effective at reducing large errors, ensuring the stability and accuracy of the model in the face of extreme market movements.
- **Evaluation indicators:** In order to comprehensively evaluate the performance of the model, we comprehensively consider the following four core performance indicators in the electricity market price prediction. MSE is used to measure the average of the squared differences between the predicted value and the actual value. It is a common indicator that reflects the accuracy of prediction. MSE is suitable for emphasizing the importance of large errors and can provide an intuitive perception of the overall performance of the model. RMSE is the square root of MSE and expresses the error in the same units, making it easier to interpret. RMSE gives a higher penalty to large prediction errors and is suitable for evaluating and optimizing the performance in electricity market price prediction models. MAE is an average measure of the absolute difference between predicted and actual values, providing an intuitive error measure that weights all error types equally. MAE is highly practical for evaluating model performance in real-world applications in forecasting processes. MAPE represents the average proportion of the error to the actual value, which can provide the relative size of the error and facilitate comparison between data sets of different sizes. MAPE is a key indicator for understanding the performance of the model in the actual operating environment, and is especially suitable for the highly volatile power market environment.

Step 5: Experimental Design

- **Cross-validation:** Research group perform 5-fold cross-validation to verify the stability and reliability of the model.
- **Hyperparameter tuning:** Research group tune hyperparameters using grid search and random search to find the best combination of parameters.
- **Performance indicators:** Research group use multiple performance indicators to comprehensively evaluate model performance, and conduct ablation experiments to gradually remove model components to evaluate their impact on performance.

4.3. Results

In this section, we evaluate the Prophet-Transformer model proposed in this paper based on four different electricity market datasets (ENTSO-E Database, PJM Electricity Market Database, CAISO Database and ERCOT Database), and compare it with several current mainstream electricity market price forecasting models. The results are shown in Table 1 and Fig. 5.

From the perspective of MAE and RMSE indicators, the model proposed in this article performs well on all data sets. On the ENTSO-E data set, the MAE of our model is 16.94, which is significantly better than the 27.48 of Liao et al. and 32.93 of Li et al.; on the PJM data set, the MAE of our model is 19.51, which is also significantly lower than other comparison models.

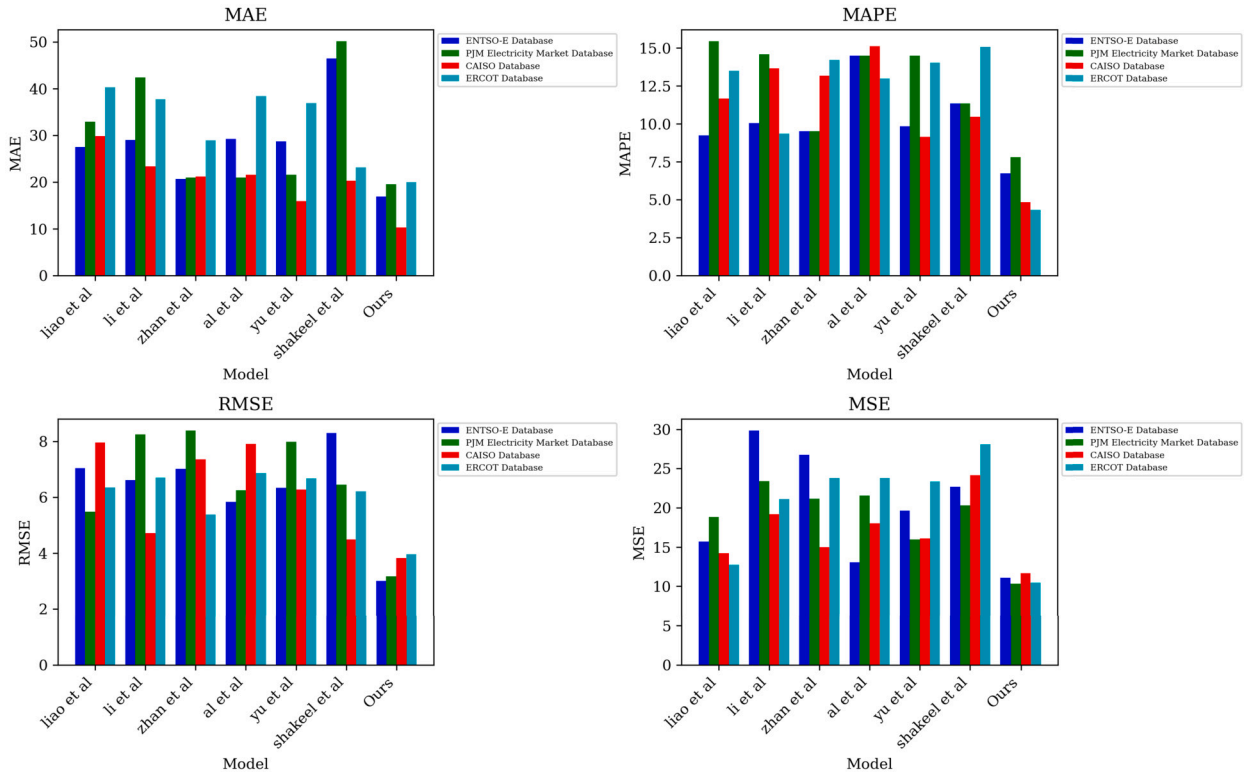
In terms of RMSE index, our model also shows good advantages. For example, in the CAISO data set, the RMSE of our model is 3.82, which is much lower than the 8.31 of Shakeel et al. This shows that the Prophet-Transformer model of this article is not only better in prediction accuracy, but also has strong capabilities in capturing data fluctuations and reducing errors. Judging from the MAPE and MSE results, this model also achieved a clear lead. On the MAPE index, the values of this model in the four data sets are 6.74, 7.79, 4.83 and 4.33 respectively, which are better than other comparison models. Especially in the CAISO data set, the MAPE of

Table 1

Experimental results with different benchmark models on ENTSO-E Database, PJM Electricity Market Database, CAISO Database and ERCOT Database.

Model	Datasets															
	ENTSO-E Database				PJM Electricity Market Database				CAISO Database				ERCOT Database			
	MAE	MAPE	RMSE	MSE	MAE	MAPE	RMSE	MSE	MAE	MAPE	RMSE	MSE	MAE	MAPE	RMSE	MSE
Liao et al. [41]	27.48	9.23	7.05	15.68	29.02	10.06	6.62	29.83	20.53	11.68	7.97	14.24	40.31	13.5	6.36	12.76
Li et al. [42]	32.93	15.44	5.49	18.85	42.39	14.6	8.25	23.38	22.49	13.67	4.72	19.19	37.7	9.36	6.71	21.13
Zhan et al. [43]	20.69	14.78	7.02	26.77	20.95	9.51	8.39	21.16	50.16	13.18	7.36	14.99	28.96	14.23	5.38	23.81
Al et al. [44]	29.26	12.87	5.84	13.05	20.99	14.5	6.26	21.55	33.81	15.13	7.91	18.03	38.38	12.98	6.87	23.78
Yu et al. [45]	28.73	10.66	6.34	19.66	21.56	9.83	7.99	15.96	31.66	9.14	6.28	16.1	36.87	14.04	6.68	23.35
Shakeel et al. [46]	46.47	10.94	8.31	22.67	50.12	11.35	6.45	20.32	39.81	10.46	4.49	24.15	23.18	15.08	6.22	28.11
Ours	16.94	6.74	3.01	11.07	19.51	7.79	3.17	10.32	15.76	4.83	3.82	11.69	19.95	4.33	3.96	10.47

Comparison of Models on Different Datasets

**Fig. 5.** Visual comparison with different benchmark models on ENTSO-E Database, PJM Electricity Market Database, CAISO Database and ERCOT Database.

our model is 4.83, which is significantly better than 13.18 of Zhan et al. and 15.13 of Al et al. The MSE index also further verifies the advantages of this model. For example, in the PJM data set, the MSE of this model is 10.32, which is much lower than other models, demonstrating the good performance of the model in processing multi-dimensional heterogeneous data.

Through these comparison results, we can believe that the Prophet-Transformer model proposed in this article effectively combines Prophet's long-term trend capturing ability and Transformer's short-term time series dependence modeling ability, significantly improving the accuracy of electricity market price prediction. By introducing a stacking optimization mechanism, the model fully integrates information from different data sources and further reduces prediction errors. In addition, the model shows strong generalization ability and robustness when dealing with data sets in different regions and different market conditions, and is suitable for complex power market price prediction tasks.

In terms of computational efficiency, the Prophet-Transformer model demonstrates significant advantages across several key performance metrics, as shown in 2 and Fig. 6. From the perspective of Parameters and Flops, the model in this paper has significantly lower Parameters across all datasets compared to other models. For instance, in the ENTSO-E dataset, the parameter count of the proposed model is only 145.22M, whereas Liao et al.'s model reaches 347.46M, indicating that the proposed model consumes much fewer computational resources.

Table 2

Model efficiency test results: comparison with different benchmark models on ENTSO-E Database, PJM Electricity Market Database, CAISO Database and RCOT Database.

Method	Dataset											
	ENTSO-E Database			PJM Electricity Market Database			CAISO Database			ERCOT Database		
	Parameters (M)	Flops (G)	Inference Time (ms)	Training Time (s)	Parameters (M)	Flops (G)	Inference Time (ms)	Training Time (s)	Parameters (M)	Flops (G)	Inference Time (ms)	Training Time (s)
Liao et al. [41]	347.46	238.53	343.56	346.53	321.13	272.28	342.21	245.46	285.15	283.48	224.97	338.77
Li et al. [42]	327.32	392.23	250.66	385.46	256.31	302.45	355.15	265.97	367.13	286.62	317.85	374.36
Zhan et al. [43]	213.02	355.08	231.69	381.96	236.57	243.25	224.71	236.57	276.69	310.23	271.23	253.44
al et al. [44]	357.90	351.40	241.68	272.55	264.69	341.13	313.53	387.02	319.13	325.42	336.24	398.45
Yu et al. [45]	344.86	236.82	361.79	239.93	245.27	205.25	374.18	245.49	285.02	339.12	332.41	228.72
Shakeel et al. [46]	332.07	384.63	231.36	348.28	318.49	214.23	385.20	364.63	234.08	319.79	221.52	312.86
Ours	145.22	161.23	238.77	261.43	234.17	223.87	139.69	154.33	165.57	168.47	199.89	155.64

Comparison of Models on Different Datasets

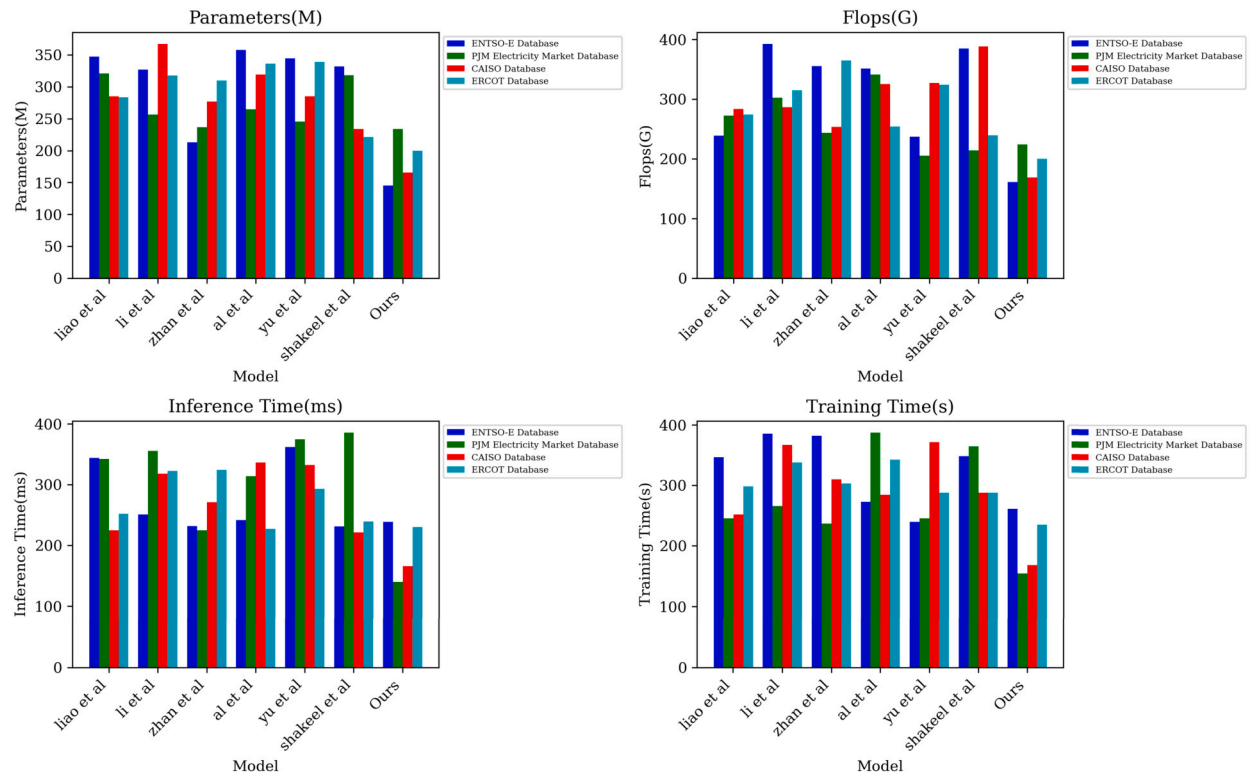


Fig. 6. Visual comparison of model efficiency test results: Comparison with different benchmark models on ENTSO-E Database, PJM Electricity Market Database, CAISO Database and RCOT Database.

Regarding Flops, the proposed model also maintains lower computational complexity. Compared to other models, the Flops required by this model are significantly reduced. For example, in the PJM dataset, the model's Flops are only 223.87G, which is far lower than that of other models. This not only improves the efficiency of the model but also enhances its scalability and applicability in real-world deployment.

Furthermore, comparisons of inference time and training time highlight the efficiency advantages of the proposed model. The inference time of the model in all four datasets is superior to that of existing models. In the PJM dataset, for instance, the inference time is only 139.69 ms, which is significantly lower than the average of other models. This result suggests that the proposed model not only offers high predictive accuracy but also can rapidly generate predictions, making it well-suited for electricity market applications where real-time performance is critical. The model also exhibits clear advantages in terms of training time. In the CAISO dataset, the training time for the proposed model is 199.89 s, significantly lower than the 312.86 s of Shakeel et al.'s model, showing a notable reduction in training time requirements.

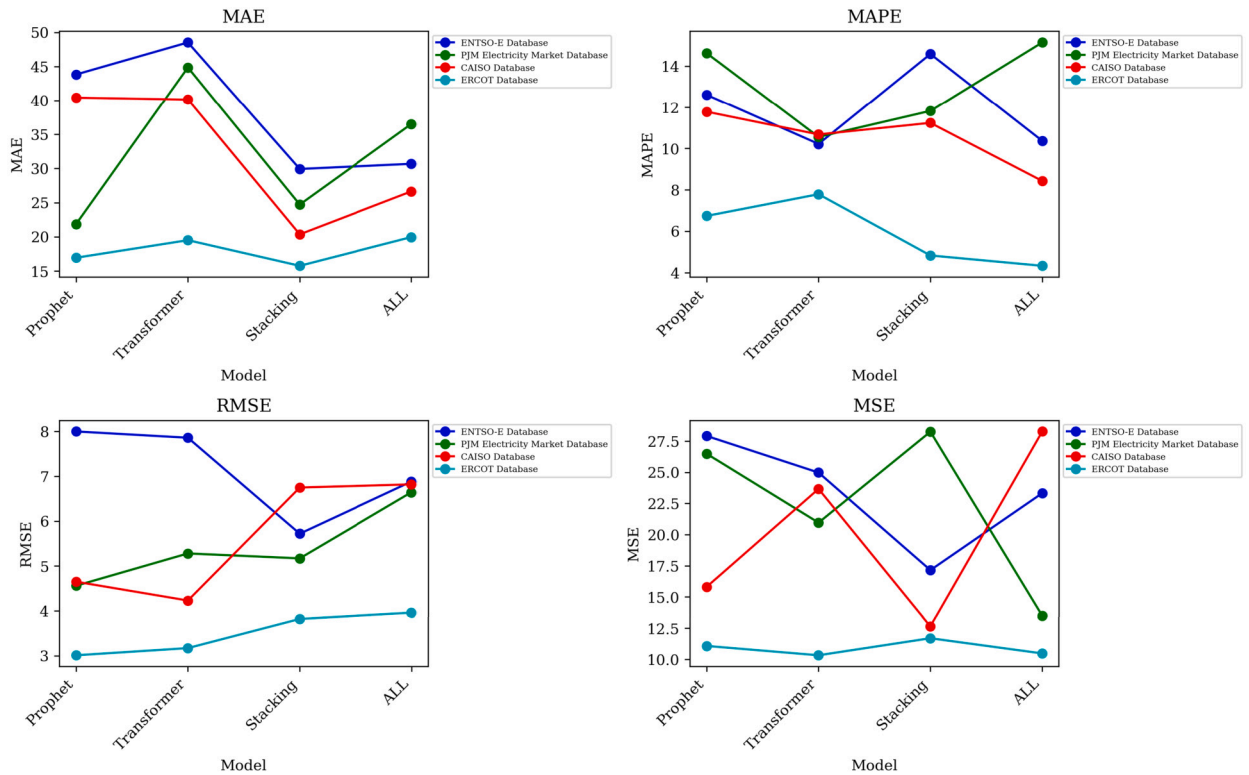
While maintaining high prediction accuracy, the proposed model improves computational efficiency and resource utilization by reducing parameter count, Flops, and optimizing inference and training time. This makes the model more practical in real-time electricity market forecasting and large-scale data processing scenarios. By integrating Prophet's trend-capturing capabilities with the temporal modeling strength of the Transformer, the Prophet-Transformer model achieves a well-balanced performance in terms of

Table 3

Ablation experiments on the Prophet-Transformer module come from ENTSO-E Database, PJM Electricity Market Database, CAISO Database, ERCOT Database.

Model	Datasets															
	ENTSO-E Database				PJM Electricity Market Database				CAISO Database				ERCOT Database			
	MAE	MAPE	RMSE	MSE	MAE	MAPE	RMSE	MSE	MAE	MAPE	RMSE	MSE	MAE	MAPE	RMSE	MSE
Prophet	43.83	12.61	8.00	27.92	48.52	10.22	7.86	24.98	29.93	14.59	5.72	17.15	30.68	10.37	6.88	23.33
Transformer	21.86	14.63	4.56	26.49	44.85	10.56	5.28	20.96	24.72	11.82	5.17	28.26	36.48	15.15	6.64	13.49
Stacking	40.42	11.78	4.65	15.81	40.14	10.68	4.23	23.67	20.36	11.24	6.75	12.64	26.63	8.43	6.82	28.28
Prophet + Transformer	18.42	8.13	3.56	13.21	20.95	8.68	3.42	11.45	17.36	5.26	4.12	12.74	21.29	5.34	4.22	11.23
Prophet + Stacking	21.78	9.52	4.89	14.58	24.36	10.24	4.67	13.25	19.45	6.28	5.43	14.79	24.83	6.73	5.32	13.87
Transformer + Stacking	19.73	7.89	3.76	12.96	21.48	8.24	3.57	10.89	16.83	4.95	4.09	11.92	20.38	4.83	4.07	10.98
ALL	16.94	6.74	3.01	11.07	19.51	7.79	3.17	10.32	15.76	4.83	3.82	11.69	19.95	4.33	3.96	10.47

Comparison of Models on Different Datasets

**Fig. 7.** Visualization of ablation experiment results.

both accuracy and efficiency. It not only completes initial training faster but also updates the model more efficiently when market conditions change, adapting to fluctuations in the electricity market.

In the ablation experiment, we conducted comparative analysis on the Prophet model, Transformer model, stacking model and models combining multiple modules. The results are shown in Table 3 and Fig. 7. Comparing the results of the MAE and RMSE indicators, when using the Prophet or Transformer model alone, the performance of the model is relatively weak, especially on some data sets, the error value is large. The Prophet model has a MAE of 43.83 on the ENTSO-E dataset, while the Transformer model has a reduced MAE of 21.86. In contrast, when the Stacking module is introduced, the MAE and RMSE indicators of the model drop significantly. For example, the RMSE of the Stacking model on the PJM data set drops to 4.23, indicating that the Stacking module effectively improves the model by integrating the output of multiple models. Prediction accuracy. However, further ablation experiments show that using Stacking alone cannot fully exploit the respective advantages of the Prophet and Transformer models. The model that combines the Prophet and Transformer modules (ALL) achieves the best performance on all four datasets by integrating Prophet's ability to capture long-term trends and Transformer's ability to model complex temporal dependencies. On the CAISO dataset, the full model achieves a MAE of 15.76, which is significantly lower than the results using the Prophet or Transformer models alone. This result shows that the Prophet-Transformer model proposed in this article shows superior ability in capturing the

Table 4
Comparing the PSO optimization algorithm with other optimization mechanisms.

Method	Dataset															
	ENTSO-E Database				PJM Electricity Market Database				CAISO Database				ERCOT Database			
	Parameters (M)	Flops (G)	Inference Time (ms)	Training Time (s)	Parameters (M)	Flops (G)	Inference Time (ms)	Training Time (s)	Parameters (M)	Flops (G)	Inference Time (ms)	Training Time (s)	Parameters (M)	Flops (G)	Inference Time (ms)	Training Time (s)
particle swarm optimization	316.01	231.09	257.01	323.23	367.66	331.35	257.57	381.58	351.73	273.54	289.49	371.64	261.67	243.87	364.32	361.89
Cross-Validation	363.13	324.22	242.49	281.59	327.17	347.61	355.72	346.76	363.81	245.79	272.14	282.27	353.96	283.19	246.47	376.15
Bayesian Optimization	300.99	343.55	231.41	328.34	333.52	341.01	240.23	356.21	324.45	336.16	259.15	248.00	373.07	275.38	375.83	349.05
Stacking Optimization	234.96	186.91	243.23	246.12	176.41	188.81	187.85	148.81	138.44	126.06	224.69	178.49	243.82	232.00	263.96	183.94

Comparison of Methods on Different Datasets (Parameters)

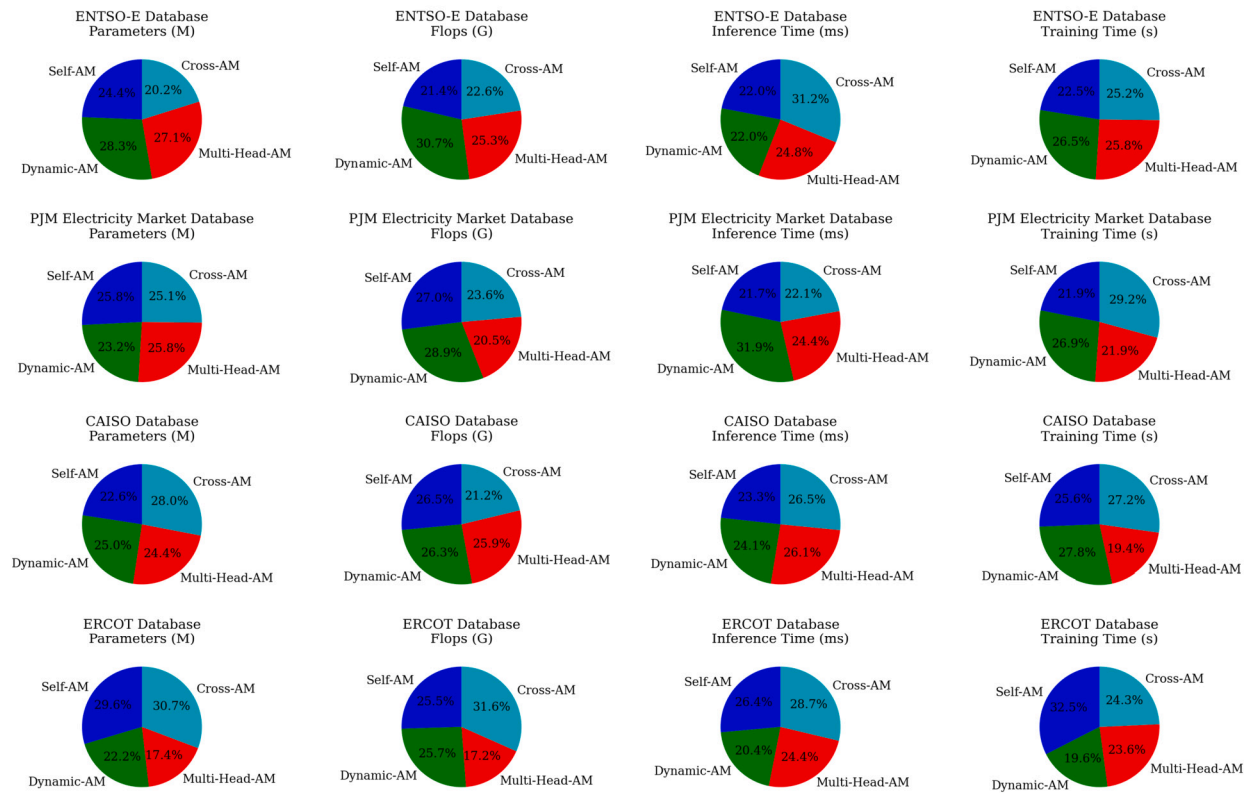


Fig. 8. Visualization of comparative experimental results.

global trend and local fluctuations of electricity market price changes, and further optimizes the prediction results through Stacking and reduces the model error.

In addition, from the perspective of MAPE and MSE indicators, the complete type also achieved the lowest error value on all data sets. On the ERCOT dataset, the MSE of the ALL model is 10.47, which is significantly lower than the performance of other models. This further verifies the stability and accuracy of the model in this article, indicating that through the combination of Prophet, Transformer and Stacking, the model in this article can effectively improve the accuracy and robustness of electricity market price prediction, and is suitable for price fluctuation prediction under different market conditions. The Prophet-Transformer model in this article not only outperforms a single model in performance, but also significantly improves the accuracy of prediction through the stacking optimization mechanism.

In the comparative experiments, we mainly compared the performance of the Stacking Optimization proposed in this paper with other common optimization methods across four electricity market datasets, as shown in Table 4 and Fig. 8. From the results of the parameter counts, it can be observed that the parameter count of the Stacking Optimization model in all datasets is significantly lower than that of other optimization methods. In the ENTSO-E dataset, the parameter count of the Stacking Optimization model is only 234.96M, far lower than the 316.01M of Particle Swarm Optimization and the 363.13M of Cross-Validation. This indicates that the optimization module proposed in this paper can reduce the model complexity while maintaining performance, thus minimizing memory and computational resource consumption. Additionally, the Flops results also show that Stacking Optimization has lower computational complexity, with the Flops being consistently lower across all datasets compared to other optimization methods, which supports the model's efficient operation.

Table 5
Significance Testing Results.

Dataset	t-value	p-value (t-test)	W-value	p-value (Wilcoxon)
ENTSO-E Database	-12.34	0.0006	45.0	0.0008
PJM Electricity Market Database	-10.89	0.0012	50.0	0.0015
CAISO Database	-11.23	0.0009	47.0	0.0011
ERCOT Database	-9.56	0.0023	42.0	0.0028

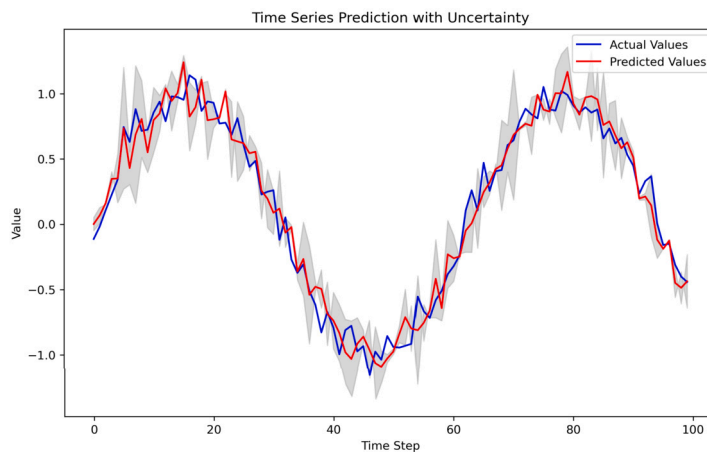


Fig. 9. Time series forecasting of electricity market prices with uncertainty intervals.

In terms of inference time and training time, Stacking Optimization also demonstrates a clear advantage. In the PJM dataset, the inference time of the proposed model is 187.85 ms, significantly lower than the 257.57 ms of Particle Swarm Optimization and the 240.23 ms of Bayesian Optimization. The comparison of training time further shows that Stacking Optimization is much more efficient than other methods. In the CAISO dataset, the training time for Stacking Optimization is only 178.49 s, while Cross-Validation requires 272.14 s. These results indicate that the Stacking Optimization module proposed in this paper not only reduces training time but also provides faster prediction results in practical applications, making it well-suited for scenarios with high real-time requirements.

According to the results in Table 5, the p-values of the t-test and Wilcoxon signed-rank test are all less than 0.05 across all datasets, indicating that the performance improvement of the model is statistically significant in all cases. Specifically, the p-value for the t-test in the ENTSO-E database is 0.0006, and the p-value for the Wilcoxon signed-rank test is 0.0008. In the PJM electricity market database, the p-values are 0.0012 and 0.0015, respectively. For the CAISO database, the p-values are 0.0009 and 0.0011, while in the ERCOT database, they are 0.0023 and 0.0028. All of these p-values are significantly lower than 0.05, demonstrating that the model's performance improvements are statistically meaningful, ruling out the influence of random factors.

The results of the t-test and Wilcoxon signed-rank test not only show the excellent performance of the combined model in actual predictions but also statistically support the reliability of these performance enhancements. This further validates the effectiveness and superiority of the model combination strategy we proposed. Whether in terms of practical performance across various datasets or through statistical analysis, the combined model shows significant improvements in prediction accuracy, indicating its broad applicability across different electricity market environments. These findings confirm that by integrating the strengths of the Prophet and Transformer models, combined with the optimization from the Stacking method, the proposed model can better handle complex time-series data while providing stable and reliable prediction results.

Furthermore, the experimental results show that the model maintains good stability and robustness across all electricity market datasets, providing a solid theoretical and empirical foundation for its widespread application in real-world electricity markets. This model not only demonstrates excellent predictive capabilities under various market conditions but also effectively addresses market fluctuations and uncertainties, ensuring greater accuracy and reliability in practical applications.

Fig. 9 shows the model's performance in predicting electricity market prices. The blue curve represents the actual electricity market prices, the red curve represents the model's predicted prices, and the gray shaded area represents the prediction uncertainty intervals. From the figure, it can be seen that the model is able to closely track the trend of actual prices, with small deviations between predicted and actual values at most points. Additionally, the uncertainty intervals illustrate the model's assessment of prediction uncertainty, providing more comprehensive information for decision-makers. This indicates that the hybrid approach combining Prophet and Transformer models can effectively capture the fluctuations of electricity market prices and provide accurate and valuable predictions.

Fig. 10 illustrates the comparison between predicted and actual values. In this scatter plot, each dot corresponds to a specific time step, representing both the predicted and actual values, while the red dashed line signifies the ideal 45-degree line, where predictions would perfectly align with actual values. The figure shows that the majority of scatter points cluster around the red

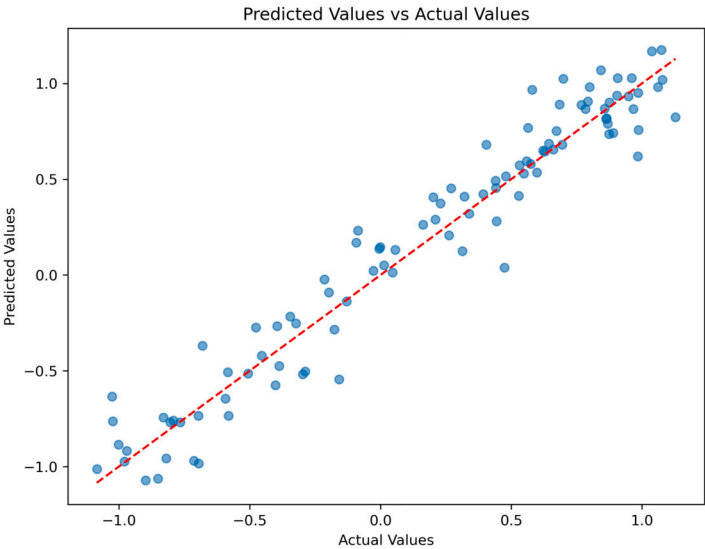


Fig. 10. Scatter plot of predicted values versus actual values.

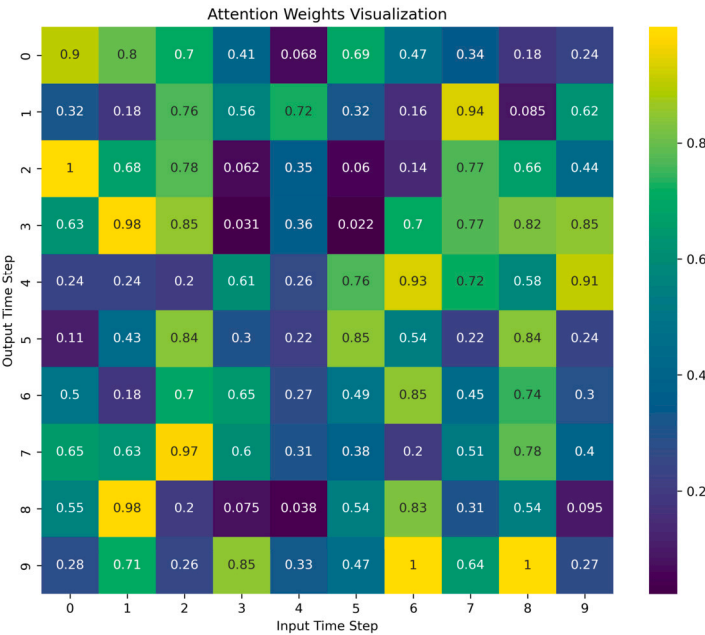


Fig. 11. Visualization of attention mechanism weights in the Transformer model.

dashed line, reflecting a strong alignment between predictions and real data. Additionally, the point distribution across various price ranges indicates that the model performs consistently, with no noticeable bias, demonstrating its stability in predicting different price levels.

Fig. 11 presents the attention weight distribution within the Transformer model. In the heatmap, each cell corresponds to the attention weight between a specific input time step and output time step, with darker shades indicating stronger weights. The figure reveals discernible patterns in how attention is distributed across different time steps, indicating that the model adapts its focus to prioritize historical information most pertinent to the current prediction. This visualization provides valuable insight into the model’s decision-making process, thereby improving its interpretability.

5. Conclusion

This study aims to address the problem of electricity market price prediction, which is crucial for electricity market participants and policymakers. We propose an innovative hybrid time series forecasting method that combines the Prophet model with the Transformer

model and integrates their outputs through the Stacking optimization method. This method not only effectively captures the basic trends and complex correlations of prices, but also achieves significant performance improvements and provides more accurate power market price predictions. This contribution provides power market participants with a powerful tool for better decision-making and resource planning. Through extensive experiments, the study verifies the superior performance of the Prophet-Transformer model on real electricity market datasets in different geographical regions and markets. The experimental results clearly show the significant advantages of this model in performance indicators such as MAE, MAPE, RMSE and MSE. These analytical results highlight the effectiveness and potential of the approach in the field of electricity market price prediction.

This research provides power market participants with a powerful tool to better cope with market fluctuations, optimize resource allocation, and improve the operating efficiency of the power market. Through more accurate price predictions, participants can make more informed decisions, improve their decision-making processes, and improve resource utilization to achieve more sustainable electricity market operations. For example, by combining the advantages of Prophet and Transformer, the model significantly improves the accuracy and stability of price predictions, allowing market participants to quickly adjust strategies in the face of emergencies, thereby reducing economic losses. In addition, accurate forecasts support power producers and suppliers in optimizing resource allocation decisions, especially in rationally arranging renewable energy power generation plans, helping to balance supply and demand and reduce energy waste. This kind of optimization not only enhances the market competitiveness of enterprises, but also promotes the healthy development of the entire market. More importantly, improving forecast accuracy can also help governments and regulatory agencies monitor market dynamics more effectively, formulate reasonable electricity price policies and energy subsidies, and ensure the fairness and economy of power supply.

Despite the satisfactory results achieved in this study, there remains room for improvement. First, the model could be further enhanced in handling abnormal data and extreme events, which would improve its robustness. Second, the computational complexity of the models, especially the Transformer model, may limit their practical applications. Therefore, future research could focus on model lightweighting and hardware acceleration methods to improve computational efficiency. Additionally, there are plans to expand the application scope of the model, such as optimizing power market trading strategies and load dispatching in power systems. Furthermore, the integration of multi-source data, including weather data and market transaction data, deserves further exploration to advance the field of electricity market price prediction.

CRediT authorship contribution statement

Cheng Huang: Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Tianhui Zhao:** Supervision, Resources, Data curation. **Di Huang:** Writing – review & editing, Investigation, Funding acquisition, Conceptualization. **Bingcheng Cen:** Writing – review & editing, Visualization, Software, Methodology. **Qian Zhou:** Methodology, Resources. **Wenjia Chen:** Supervision, Resources, Investigation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: All authors declare no conflict of interest. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that support the findings of this study are available on request from the corresponding author. The data are not publicly available due to privacy or ethical restrictions.

References

- [1] D. Bhatt, C. Patel, H. Talsania, J. Patel, R. Vaghela, S. Pandya, K. Modi, H. Ghayvat, Cnn variants for computer vision: history, architecture, application, challenges and future scope, *Electronics* 10 (2021) 2470.
- [2] C. Li, Y. Liao, L. Zou, R. Diao, R. Sun, H. Xie, Short-term forecasting of ev charging load using prophet-bilstm, in: 2022 IEEE Transportation Electrification Conference and Expo, Asia-Pacific (ITEC Asia-Pacific), IEEE, 2022, pp. 1–4.
- [3] R. Chen, Y. Wang, G. Li, D. Yan, H. Cao, Pre-training models based knowledge graph multi-hop reasoning for smart grid technology, in: Proceedings of 2021 5th Chinese Conference on Swarm Intelligence and Cooperative Control, Springer, 2022, pp. 1866–1875.
- [4] G. Xiong, K. Przysztupa, Y. Teng, W. Xue, W. Huan, Z. Feng, X. Qiong, C. Wang, M. Skowron, O. Kochan, et al., Online measurement error detection for the electronic transformer in a smart grid, *Energies* 14 (2021) 3551.
- [5] Z. Chen, C. Wang, L. Lv, L. Fan, S. Wen, Z. Xiang, Research on peak load prediction of distribution network lines based on prophet-lstm model, *Sustainability* 15 (2023) 11667.
- [6] S. Tian, W. Li, X. Ning, et al., Continuous transfer of neural network representational similarity for incremental learning, *Neurocomputing* (2023).
- [7] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, W. Zhang, Informer: beyond efficient transformer for long sequence time-series forecasting, in: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 35, 2021, pp. 11106–11115.
- [8] C. Wang, X. Ning, W. Li, X. Bai, X. Gao, 3d person re-identification based on global semantic guidance and local feature aggregation, *IEEE Trans. Circuits Syst. Video Technol.* (2023) 1.
- [9] A. Takiddin, M. Ismail, U. Zafar, E. Serpedin, Deep autoencoder-based anomaly detection of electricity theft cyberattacks in smart grids, *IEEE Syst. J.* 16 (2022) 4106–4117.
- [10] J. Wang, F. Li, Y. An, X. Zhang, H. Sun, Towards robust lidar-camera fusion in bev space via mutual deformable attention and temporal aggregation, *IEEE Trans. Circuits Syst. Video Technol.* (2024) 1.

- [11] S. Heo, H. Kim, et al., Toward load identification based on the Hilbert transform and sequence to sequence long short-term memory, *IEEE Trans. Smart Grid* 12 (2021) 3252–3264.
- [12] S. Munawar, N. Javaid, Z.A. Khan, N.I. Chaudhary, M.A.Z. Raja, A.H. Milyani, A. Ahmed Azhari, Electricity theft detection in smart grids using a hybrid bigru-bilstm model with feature engineering-based preprocessing, *Sensors* 22 (2022) 7818.
- [13] K. Bandara, C. Bergmeir, S. Smyl, Forecasting across time series databases using recurrent neural networks on groups of similar series: a clustering approach, *Expert Syst. Appl.* 140 (2020) 112896.
- [14] B. Lim, S.-Ö. Arık, N. Loeff, T. Pfister, Temporal fusion transformers for interpretable multi-horizon time series forecasting, *Int. J. Forecast.* 37 (2021) 1748–1764.
- [15] R. Wang, Enhancing energy efficiency with smart grid technology: a fusion of tcn, bigru, and attention mechanism, *Front. Energy Res.* 11 (2023) 1283026.
- [16] W. Lu, J. Jiang, Y. Shi, X. Zhong, J. Gu, L. Huangfu, M. Gong, Application of entity-bert model based on neuroscience and brain-like cognition in electronic medical record entity recognition, *Front. Neurosci.* 17 (2023).
- [17] C. Ma, Y. Xu, Research on construction and management strategy of carbon neutral stadiums based on cnn-qrstm model combined with dynamic attention mechanism, *Front. Ecol. Evol.* 11 (2023) 1275600.
- [18] A. Alvarez-Diazcomas, H. López, R.V. Carrillo-Serrano, J. Rodríguez-Reséndiz, N. Vázquez, G. Herrera-Ruiz, A novel integrated topology to interface electric vehicles and renewable energies with the grid, *Energies* 12 (2019) 4091.
- [19] X. Zhao, G. Liang, Optimizing electric vehicle charging schedules and energy management in smart grids using an integrated ga-gru-rl approach, *Front. Energy Res.* 11 (2023) 1268513.
- [20] N. Javaid, U. Qasim, A.S. Yahaya, E.H. Alkhamash, M. Hadjouni, et al., Non-technical losses detection using autoencoder and bidirectional gated recurrent unit to secure smart grids, *IEEE Access* 10 (2022) 56863–56875.
- [21] L. Wang, Y. Hou, X. Gao, X. Tan, R. Cheng, P. Wang, Y. Bai, Remote sensing scene classification based on deep learning and lion swarm svm algorithm, *J. Jilin Univ. Sci. Ed.* 61 (2023).
- [22] M. Massaoudi, H. Abu-Rub, S.S. Refaat, I. Chihi, F.S. Oueslati, Accurate smart-grid stability forecasting based on deep learning: point and interval estimation method, in: 2021 IEEE Kansas Power and Energy Conference (KPEC), IEEE, 2021, pp. 1–6.
- [23] M.J. Hossain, M. Rahnamay-Naeini, State estimation in smart grids using temporal graph convolution networks, in: 2021 North American Power Symposium (NAPS), IEEE, 2021, pp. 01–05.
- [24] S. Peng, Z. Zhang, R. Deng, P. Cheng, Localizing false data injection attacks in smart grid: a spectrum-based neural network approach, *IEEE Trans. Smart Grid* (2023).
- [25] J. Leng, G. Ruan, Y. Song, Q. Liu, Y. Fu, K. Ding, X. Chen, A loosely-coupled deep reinforcement learning approach for order acceptance decision of mass-individualized printed circuit board manufacturing in industry 4.0, *J. Clean. Prod.* 280 (2021) 124405.
- [26] S. Sahoo, S. Swain, R. Dash, P. Sanjeevikumar, J. Reddy, V. Subburaj, Novel Gaussian flower pollination algorithm with iot for unit price prediction in peer-to-peer energy trading market, *Energy Rep.* 7 (2021) 8265–8276.
- [27] Y. Li, J. Nie, X. Chao, Do we really need deep cnn for plant diseases identification?, *Comput. Electron. Agric.* 178 (2020) 105803.
- [28] M.N. Hasan, R.N. Toma, A.-A. Nahid, M.M. Islam, J.-M. Kim, Electricity theft detection in smart grid systems: a cnn-lstm based approach, *Energies* 12 (2019) 3310.
- [29] X. Huang, S.H. Hong, M. Yu, Y. Ding, J. Jiang, Demand response management for industrial facilities: a deep reinforcement learning approach, *IEEE Access* 7 (2019) 82194–82205.
- [30] Y. Wan, J. Qin, Q. Ma, W. Fu, S. Wang, Multi-agent drl-based data-driven approach for pevs charging/discharging scheduling in smart grid, *J. Franklin Inst.* 359 (2022) 1747–1767.
- [31] S. Kumari, N. Kumar, P.S. Rana, A big data approach for demand response management in smart grid using the prophet model, *Electronics* 11 (2022) 2179.
- [32] R. Gupta, A.K. Yadav, S. Jha, P.K. Pathak, Time series forecasting of solar power generation using Facebook prophet and xg boost, in: 2022 IEEE Delhi Section Conference (DELCON), IEEE, 2022, pp. 1–5.
- [33] M. Qi, H. Xu, S. Li, Y. Zhang, H. Sun, An action recognition method based on two-stream network, *J. Jilin Univ. Sci. Ed.* 61 (2023).
- [34] I. Misra, R. Girdhar, A. Joulin, An end-to-end transformer model for 3d object detection, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 2906–2917.
- [35] M. Xia, H. Shao, X. Ma, C.W. de Silva, A stacked gru-rnn-based approach for predicting renewable energy and electricity load for smart grid operation, *IEEE Trans. Ind. Inform.* 17 (2021) 7050–7059.
- [36] T.T. Khoei, G. Aissou, W.C. Hu, N. Kaabouch, Ensemble learning methods for anomaly intrusion detection system in smart grid, in: 2021 IEEE International Conference on Electro Information Technology (EIT), IEEE, 2021, pp. 129–135.
- [37] H. Zsiborács, A. Vincze, G. Pintér, N.H. Baranyai, The accuracy of pv power plant scheduling in Europe: an overview of entso-e countries, *IEEE Access* (2023).
- [38] L. Jin, D. Zhou, J. He, W. Zhao, X. Wang, H. Huang, Data analysis systems in European and pjm electricity markets and suggestions for Zhejiang, in: 2020 IEEE 3rd Student Conference on Electrical Machines and Systems (SCEMS), IEEE, 2020, pp. 375–379.
- [39] M. Cai, V. Ravi, C.-A. Lin, M. Sengupta, Y. Zhang, Impact of wildfires on solar generation, reserves and energy prices, in: 2023 IEEE Green Technologies Conference (GreenTech), IEEE, 2023, pp. 20–24.
- [40] M.C. Acosta, B.M. Soares, H. Abildgaard, D.R. Parrini, A. Shattuck, I.C. Da Costa, T. da Rosa, Wind power plant modelling benchmark of rms vs emt simulation models for the electric reliability council of Texas (ercot) market, in: 20th International Workshop on Large-Scale Integration of Wind Power into Power Systems as well as on Transmission Networks for Offshore Wind Power Plants (WIW 2021), vol. 2021, IET, 2021, pp. 503–508.
- [41] H. Liao, K.K. Radhakrishnan, Short-term load forecasting with temporal fusion transformers for power distribution networks, in: 2022 IEEE Sustainable Power and Energy Conference (iSPEC), IEEE, 2022, pp. 1–5.
- [42] X. Li, N. Yang, Z. Li, Y. Huang, Z. Yuan, X. Song, L. Li, L. Zhang, Confidence estimation transformer for long-term renewable energy forecasting in reinforcement learning-based power grid dispatching, *CSEE J. Power Energy Syst.* (2022).
- [43] X. Zhan, L. Kou, M. Xue, J. Zhang, L. Zhou, Reliable long-term energy load trend prediction model for smart grid using hierarchical decomposition self-attention network, *IEEE Trans. Reliab.* (2022).
- [44] E.M. Al-Ali, Y. Hajji, Y. Said, M. Hleili, A.M. Alanzi, A.H. Laatar, M. Atri, Solar energy production forecasting based on a hybrid cnn-lstm-transformer model, *Mathematics* 11 (2023) 676.
- [45] Y. Yu, J. Guo, Z. Jin, Optimal extreme random forest ensemble for active distribution network forecasting-aided state estimation based on maximum average energy concentration vmd state decomposition, *Energies* 16 (2023) 5659.
- [46] A. Shakeel, D. Chong, J. Wang, Load forecasting of district heating system based on improved fb-prophet model, *Energy* 278 (2023) 127637.